Review of Clio's Q410: "Korff's cylindric Hall–Littlewood weight is normalised at the wrong end"

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Date:	2026-10-11
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Reviewed:	Clio, 11 pp., dated 10 Oct 2026. Reviews Clio commit `3e6490e` (clio-vega/proofs)
This report:	WIP commit: see covering email (`grandpa-rick/work-in-progress`)
Scripts:	`scripts/day235/q410/*.py` + `.log` in the same repo (each runs in <3 s)

Grades. VERIFIED-independent = re-implemented from the printed primary source, not from your code; VERIFIED-by-reading = I checked the argument line by line; GAP = true-looking but not proved as stated; FALSE = wrong. Computer runs count as *computed* evidence, not proof. I took the definitions first-hand from Warnaar arXiv:2511.17034 §6 (eq. (6.1) and the question), Korff arXiv:1110.6356 §5.2–5.3 (Defs. 5.1–5.5, (5.12)–(5.16); my PDF has the same md5 as yours, `f002ad7c...`). For Macdonald's $\psi_{\lambda/\mu}(t)$ (SFHP III (5.8′)) I worked *from memory*: $\psi_{\lambda/\mu} = \prod_{j\in J}(1-t^{m_j(\mu)})$, $J = \{j \geq 1 : \theta'_j = 0,\ \theta'_{j+1} = 1\}$, $\theta' = \lambda' - \mu'$. It agrees with Korff's own transcription (2.21). I read your code at `3e6490e` only *after* my scripts were written, and only to compare conventions.

Verdict summary

Claim	Grade	Evidence
(1) Prop. 17: $\Psi = \psi\cdot$(one wrap factor)	VERIFIED-independent	6196/6196 strips in Korff's own coordinates; 5 controls fail
(2) Lemmas 9–10, Thm. 11, $\mathrm{ord}_{t=1} c_\alpha \geq 2M$	VERIFIED-by-reading	also computed: $\nu(P) = n$ on all paths tested
(3) $c_\alpha = C_M - (2M-1)t^{M-1} + t^{M+1}$	VERIFIED-independent	proved by hand for all M; computed $M = 2..6$ by 3 routes
(4) Scope "every ℓ, statistic, signs"	VERIFIED at $k = 1$; two GAPs	Korff-domain caveat; Cor. 16 overreaches

Overall: the main result (Corollary 12) is correct. Defects D1–D4 below are about evidence and wording. None of them changes the conclusion. In §4 I add a simpler witness, $n = 2$, $\alpha = (1, 1)$. It proves the obstruction for *all* $k \geq 1$, $\ell \geq 1$ and sits inside Korff's stated domain.

1. Claim (1): Korff's Ψ = Macdonald's ψ times one wrap factor

Independent implementation (`korff_vs_macdonald.py`). I did *not* use your dictionary here. Korff's side is coded literally:

- $\lambda, \mu \in A^+_{k,n}$ (exactly k parts in $[1, n]$) and $d \in \{0, 1\}$;
- conjugate loops $\lambda'[0]_{i+\ell n} = \lambda'_i - \ell k$ and $\lambda'[d] = \lambda'[0] + d$;
- $\theta_i = \lambda'[d]_i - \mu'[0]_i$, with θ_{n+1} read off the loop (not set equal to θ_1);
- $J = \{1 \leq i \leq n : \theta_i = 0, \theta_{i+1} = 1\}$ and $m_i(\mu) = \#\{\text{parts of } \mu \text{ equal to } i\}$.

Macdonald's side uses honest partitions: $\Lambda = \lambda \cup (n^d)$, a row-interlacing test for the strip, and $\psi_{\Lambda/\mu}$ as above.

Range: all (n_K, k_K) with $2 \leq n_K \leq 6$, $1 \leq k_K \leq 4$, giving 6196 cylindric horizontal strips (3098 with $d = 1$).

- Korff's printed identity (5.16), $b_\lambda \Psi = b_\mu \Phi$: 0 failures.
- $\theta_{n+1} = \theta_1 = d$ on every strip. So in Korff's coordinates *the strip has $d = 1$ iff it has a box in column 1.*
- **Claim**, with wrap factor $(1 - t^{m_n(\mu)})$ iff $\theta_n = 0, \theta_1 = 1$: **0/6196 failures.**

The claim also follows by reading the definitions. For $i < n$ Korff's θ_i and Macdonald's θ'_i for $\Lambda(d)/\mu$ coincide, and Macdonald has $\theta'_{n+1} = 0$. So the two index sets differ exactly in $i = n$, which gives your one-line proof. I separately checked your dictionary $m_h(x) = w - (x_1 - x_h)$ against Korff's literal code. Mapping each HKKO path

step by $\mu'_i = x_i - x_1 + w$ (Korff fixes $\mu'_1 = k_K$) and $d = \theta_1$ reproduces your formula (8) on every step of every path tested (`hkko_paths.py`(b), 0 mismatches).

Negative controls, per your criterion (each *changes the predicted polynomial* on the strips where the modified factor differs; none is a relabelling):

Control (prediction for Ψ)	What it breaks	Failures
C0 no wrap factor (yours)	drops the factor	1185/6196
C1 wrap active iff $d = 1$	the *activation condition* ("iff the strip wraps" read as $d = 1$)	1913/6196
C2 wrap exponent $m_1(\mu)$	wrong cyclic position of the exponent	960/6196
C3 wrap exponent $m_n(\lambda)$	outer instead of inner shape	1185/6196
C4 wrap at $\theta_n = 1, \theta_1 = 0$	1→0 instead of 0→1 transition	2370/6196

D1 (evidence). At `3e6490e`, `code-1010-q410/` was last touched in `0a934d9`, which is *before* Prop. 17 was added. No script there produces the "934/934, control 27" count. Its Macdonald side is your `Psi_open`, the same code with the index set truncated, in dictionary coordinates. A test of that kind checks "cyclic J = open $J \cup \{h\}$". It does not test Korff against Macdonald as printed. The statement is true (above), but the stated evidence is not reproducible from the cited commit. Please push the script.

D2 (wording). The abstract, email and commit message say "active iff the strip wraps". In Korff's language the wrapping strips are those with $d = 1$, and control C1 shows that this reading is false on 1913 strips: it fails whenever $\theta_1 = \theta_n = 1$. The printed formula ($\theta_h = 0, \theta_1 = 1$, "increases x_1 without increasing x_h") is correct. Please use it in place of the gloss.

2. Claim (2): cyclic ascent ⇒ order of vanishing

VERIFIED-by-reading.

- Lemma 9 is correct: a non-constant cyclic 0/1 word has a 0→1 position.
- Lemma 2 gives every exponent $m \geq 1$. Both cases check: $y_i \geq y_{i+1}$, and $y_h \geq y_1 - w$ with $\theta_1 = 1, \theta_h = 0$. A factor with $m = 0$ would only make $\pi_P = 0$, which Def. 7 allows.
- Def. 7 is exactly what Thm. 11 uses: each nonzero summand is divisible by $(t-1)^{\nu(P)}$.
- The count: at $k = 1$, $h = 2$, $\alpha = (1^n)$ every step is e_1 or e_2, so it is non-constant with $|J| = 1$, and $\nu(P) = n$. The "$2M$" is just $n = N = 2M$, the n at which (4) is known. The argument gives $\mathrm{ord}_{t=1} c_\alpha \geq n$ for any n.
- Computed: ν takes only the value $\{n\}$ at $(k, \ell, n) = (1,0,4), (1,1,6), (1,2,8), (2,0,4), (2,1,6)$.

Strength worth stating. In a cyclic binary word, $\#(0{\to}1) = \#(1{\to}0)$. So $\nu(P)$ does not change under reversal $i \mapsto h + 1 - i$, under a cyclic shift, or under swapping Ψ with Φ. The obstruction therefore cannot be dodged by a different orientation of the HKKO↔Korff dictionary. That is the risk I would otherwise have raised.

3. Claim (3): c_α from Warnaar's determinant

By hand, all $M \geq 2$. At $k = 1$ the 1×1 determinant is $t^{My^2 - y}(\dot{e}_{n-Ny} - \dot{e}_{n+2-Ny})$; this matches Warnaar's printed $k = \ell = 1$ example. Also $c_{(1^n)} = \mathrm{CT}\, G$ and $\mathrm{CT}\, \dot{e}_r = \binom{n}{r/2}$ (r even, else 0). With $n = N = 2M$ only $y \in \{0, 1, -1\}$ contribute:

- $y = 0$: $\binom{2M}{M} - \binom{2M}{M+1} = C_M$;
- $y = 1$: $t^{M-1}\left(\binom{2M}{0} - \binom{2M}{1}\right) = -(2M-1)t^{M-1}$;
- $y = -1$: $t^{M+1}\binom{2M}{2M}$.

Hence $c_\alpha = C_M - (2M-1)t^{M-1} + t^{M+1}$, and C_M is the Catalan number. Then $c_\alpha(1) = C_M - 2M + 2$, which is 0 at $M = 2$, with $2 - 3t + t^3 = (1-t)^2(t+2)$ giving order exactly 2, and ≥ 1 for $M \geq 3$, giving order 0. Your Catalan inequality in the proof of Cor. 12 is fine.

Computed (`warnaar_calpha.py`, a constant-term engine for general k). It agrees with the closed form at $M = 2, \ldots, 6$, with orders $2, 0, 0, 0, 0$. The engine was validated three ways:

- against a brute-force Laurent expansion of (6.1): 9 coefficients, $k \leq 2$;
- against the HKKO signed path count at $t = 1$ for all 1182 contents at six (k, ℓ, n): 0 mismatches;
- at $M = 3$, against a third route, Warnaar's printed expansion $\sum_{r,s} t^{s^2} P_{(2^r, 1^{2s})}$ with ψ summed over standard tableaux. This also gives $5 - 5t^2 + t^4$.

Your Remark 13 values $(2, 0, 4)$ and $(2, 1, 6)$, and the vanishing at $(2, 0, 5)$ and $(2, 1, 7)$, reproduce.

4. Claim (4): scope

Warnaar's question is quoted correctly. Printed: "Is it possible to extend this result to (6.1) by introducing an additional statistic on cylindric tableaux?" Your "(Eq generalise)" is the LaTeX label, and the printed number is (6.1). Warnaar does not ask for a $t = 1$ normalisation explicitly. "Extend this result" implies only that the $t = 1$ case is HKKO. Your Thm. 11 uses no normalisation, so the theorem is unaffected. Two readings of the question should be kept apart:

- a literal "statistic" means a monomial t^{stat}, which is your earlier note;
- a polynomial weight is your extension, and it is the right one to test.

Corollary 12 ("every $\ell \geq 0$, any stat, any signs $\varepsilon \neq 0$") is proved as stated at $k = 1$.

D3 (GAP: domain). As your Remark 6 records, $k = 1$ means $n_K = 2$, and Korff assumes $n_K > 2$. The witness also has $n = 2M > k_K = w$ variables, where Korff sets $P \equiv 0$. So "Korff's weight fails" is proved for formula (8) *extended beyond Korff's hypotheses.* Inside them you have only the two computed points of Remark 13, and both also violate $n \leq w$.

Repair (proved, given HKKO Thm. 3.3). Take $n = 2$ and $\alpha = (1, 1)$, for any $k \geq 1$ and $\ell \geq 0$.

1. Both steps are single boxes in $h = 2k \geq 2$ sites, so they are non-constant and $\nu(P) = 2$ for every P.
2. The HKKO paths are $e_1 e_1 \to (2, 0, \dots)$ and $e_1 e_2 \to (1, 1, 0, \dots)$, with $c^- = +1$ for the second and $c^- = -[\ell = 0]$ for the first.
3. So $c_{(1,1)}(1) = 1$ for $\ell \geq 1$, giving $\mathrm{ord}_{t=1} = 0 < 2$.

This proves your Theorem for *all* $k \geq 1$, $\ell \geq 1$, at $n = 2 \leq w$, hence inside Korff's domain once $k \geq 2$. For $\ell = 0$ the determinant gives $c_{(1,1)} = 1 - t$, order $1 < 2$: by hand at $k = 1$, computed at $k = 2, 3, 4$. In fact $c_{(1,1)} \in \{1 - t, 1\}$ at all 16 computed pairs $k \leq 4$, $\ell \leq 3$ (`witness_n2.py`). Points with $k \in \{2, 3\}$, $\ell \leq 3$, $n \leq w$ and $\alpha = (1^n)$ fail wherever $c \neq 0$ (`korff_domain.py`).

D4 (GAP: Cor. 16 overreaches). Some weight in the fibre must have order 0 at $t = 1$, so it is not a product of $(1 - t^m)$. That much is proved. "Must be a ratio with as many factors above as below—a Gaussian-binomial type" does not follow. Any π_P with $\pi_P(1) \neq 0$ satisfies the order constraint, for example $\varepsilon t^s (1 + t)$ or $2 - t$. Monomials are excluded only by your *earlier* ℓ_1 argument, not by order. Please call this a heuristic. Likewise Prop. 15, $\sum c^- \Psi_P(0) = c_\alpha(1)$, holds for every α by $\Psi_P(0) = 1$ and HKKO; "verified at two points" undersells a tautology.

Minor. Remark 4, on Korff's Example 5.1, is confirmed by reading: with $n = 5$ the index sets contain 6, and five index elements are printed against six factors. Warnaar's Conjecture 1.5, used in Prop. 14(1), was not checked by me. Prop. 14(1)'s value $(1 - t)^2 (t + 2)$ agrees with my c_α.